Jiakun Yan

jiakun.yan@amd.com • jiakunyan.github.io

Research Interests

My research interest lies in parallel computing, especially in **high-performance RDMA-based system design**. Currently, I am focusing on designing communication libraries and programming models for large-scale GPU and CPU clusters for HPC and AI applications. I am the main developer of *Lightweight Communication Interface (LCI)* and also contributed/contributing to *MPICH* , *HPX* , *Legion* , and *Charm++* .

Education

University of Illinois at Urbana-Champaign — **Illinois, USA**
Aug. 2020 - Dec. 2025

- Computer Science PhD program, advised by Marc Snir .
- Research around high-performance communication libraries, especially *Lightweight Communication Interface (LCI)* , investigating their support for asynchronous and multithreaded execution and their implications for emerging programming models.
- GPA: 4.0/4.0

Shanghai Jiao Tong University — **Shanghai, China**
Sep. 2016 - Jun. 2020

- Bachelor's Degree of Engineering, Dept. of Computer Science.
- Zhiyuan Honors Program of Engineering (an elite program for top 5% talented students)
- GPA: 91.88/100 | Ranking: 4^{th}/151.

University of California, Berkeley — **California, USA**
Jan. 2019 - May 2019

- Exchange student, Berkeley Global Access Discover Program, GPA: 4.0/4.0.

Experience

GPU Communication Software Research — **AMD Research**
Software System Design Engineer, working with Brad Beckmann — *Jan. 2026 -*

- Conduct research on communication libraries and programming models for large-scale GPU clusters.

CS Ph.D. Program — **UIUC**
Research Assistant, advised by Marc Snir — *Aug. 2020 - Dec. 2025*

- Benchmarked low-level network features to enable efficient multithreaded communication.
- Designed and implemented LCI, a lightweight communication interface that improved asynchronous multithreaded communication performance.
- Analyzed communication requirements of asynchronous many-task systems to identify bottlenecks and optimization opportunities.
- Optimized HPX and Charm++ runtimes by integrating advanced communication libraries, including LCI and MPI extensions, achieving significant performance improvements.

GPU Software, Legate Group — **NVIDIA**
Software Engineer Intern, working with Manolis Papadakis and Hessam Mirsadeghi — *May. 2024 - Aug. 2024*

- Performance profiling and optimization for *Legion* UCX backend.

Programming Models and Runtime Systems Group — **Argonne National Laboratory**
Research Intern, advised by Yanfei Guo — *May. 2023 - Aug. 2023*

- Designed and evaluated the MPIX Continuation Proposal in MPICH.

Programming Systems and Applications Research Group — **NVIDIA Research**
Research Intern, advised by Michael Bauer and Michael Garland — *May. 2022 - Aug. 2022*

- Designed and implemented collective communication operations in *Legion* Realm.

PASSION Lab — **Lawrence Berkeley Laboratory**
Research Assistant, advised by Aydın Buluç and Katherine Yelick — *Aug. 2019 - Jan. 2020*

- Designed Asynchronous RPC Library (ARL): a high-throughput RPC system with node-level aggregation and single-node work-stealing.
- Benchmarked RDMA vs. RPC for Implementing Distributed Data Structures.

Publication

- Minyu Cheng, **Jiakun Yan**, Marc Snir. *Multithreaded Fine-Grained Asynchronous BSP for Integer Sorting with LCI and OpenMP*, 28th Workshop on Advances in Parallel and Distributed Computational Models (IPDPS APDCM), 2026. (To appear).
- **Jiakun Yan**, Marc Snir. *LCI: a Lightweight Communication Interface for Efficient Asynchronous Multithreaded Communication*, Proceedings of the 2025 International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2025.
- **Jiakun Yan**, Marc Snir, Yanfei Guo. *Examining MPI and its Extensions for Asynchronous Multithreaded Communication*, Proceedings of the 32nd European MPI Users' Group Meeting (EuroMPI), 2025.
- **Jiakun Yan**, Hartmut Kaiser, Marc Snir. *Understanding the Communication Needs of Asynchronous Many-Task Systems – A Case Study of HPX+LCI*, preprint, 2025.
- **Jiakun Yan**, Marc Snir. *Contemplating a Lightweight Communication Interface for Asynchronous Many-Task Systems*, Workshop on Asynchronous Many-Task Systems and Applications (WAMTA), 2025.
- Gregor Daiß, Patrick Diehl, **Jiakun Yan**, John K. Holmen, Rahulkumar Gayatri, Christoph Junghans, Alexander Straub, Jeff R. Hammond, Dominic Marcello, Miwako Tsuji, Dirk Pflüger, Hartmut Kaiser. *Asynchronous-Many-Task Systems: Challenges and Opportunities – Scaling an AMR Astrophysics Code on Exascale machines using Kokkos and HPX*, International Journal of High Performance Computing Applications, 2025.
- **Jiakun Yan**, Hartmut Kaiser, and Marc Snir. *Design and Analysis of the Network Software Stack of an Asynchronous Many-task System – The LCI parcelport of HPX*, SC'23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis (SC-W), 2023.
- Benjamin Brock, Yuxin Chen, **Jiakun Yan**, John Owens, Aydın Buluç, and Katherine Yelick. *RDMA vs. RPC for Implementing Distributed Data Structures*, 2019 IEEE/ACM 9th Workshop on Irregular Applications: Architectures and Algorithms (IA^3), 2019.

Honors and Awards

- **Best Poster Award**, WAMTA24 — Feb. 2024
  A Lightweight Communication Interface for Asynchronous Many-Task Systems

Skills

- **Programming Language:** C, C++, Python, CUDA, Java, Rust, Go
- **Library & Framework:** libibverbs, libfabric, UCX, MPI, GASNet-EX, UPC++, OpenSHMEM, Argobots